

LITERATURE REVIEW — INTERVIEW PRESENTATION

AMZ Driverless

The Full Autonomous Racing System

[Your name]

Template — talking points to be edited together

THE PROBLEM

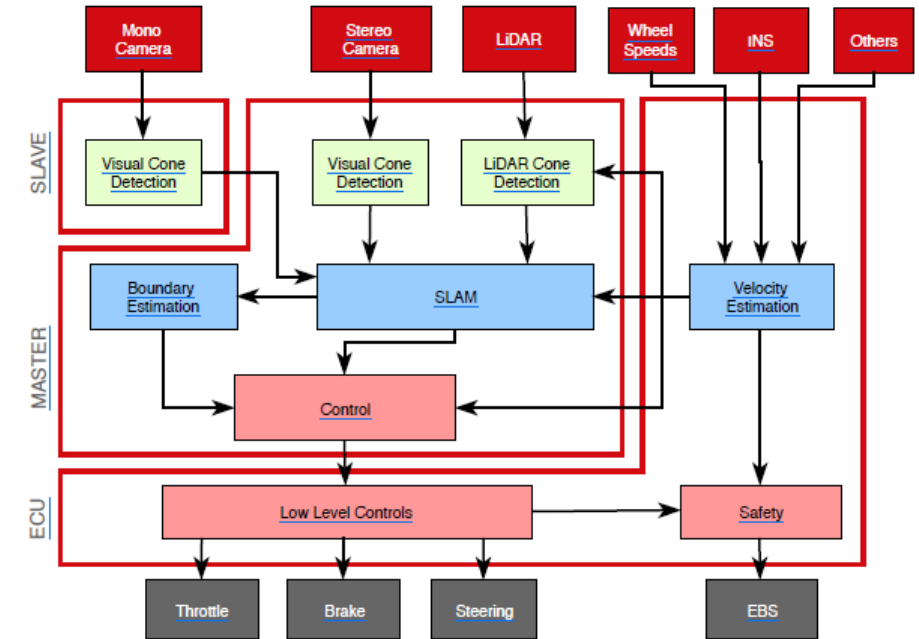
Race an Unknown Track — Fully Autonomously

- Formula Student Driverless — 10 laps, unknown track
- Boundaries = cones only (blue left / yellow right)
- All sensing + compute on-board
- Two modes: SLAM (lap 1) → Localization (laps 2–10)



Three Modules, Three Compute Units

- Perception → Motion Est. & Mapping → Control
- Slave computer: perception
- Master computer: mapping + control
- ECU: low-level control + safety (EBS)



LiDAR Pipeline

PERCEPTION · LIDAR

- Pre-processing — FoV trim + ground removal
- Cone detection — clustering + point-count filter
- Color estimation — CNN on intensity pattern

Two Things Worth Getting Right

PERCEPTION · LIDAR

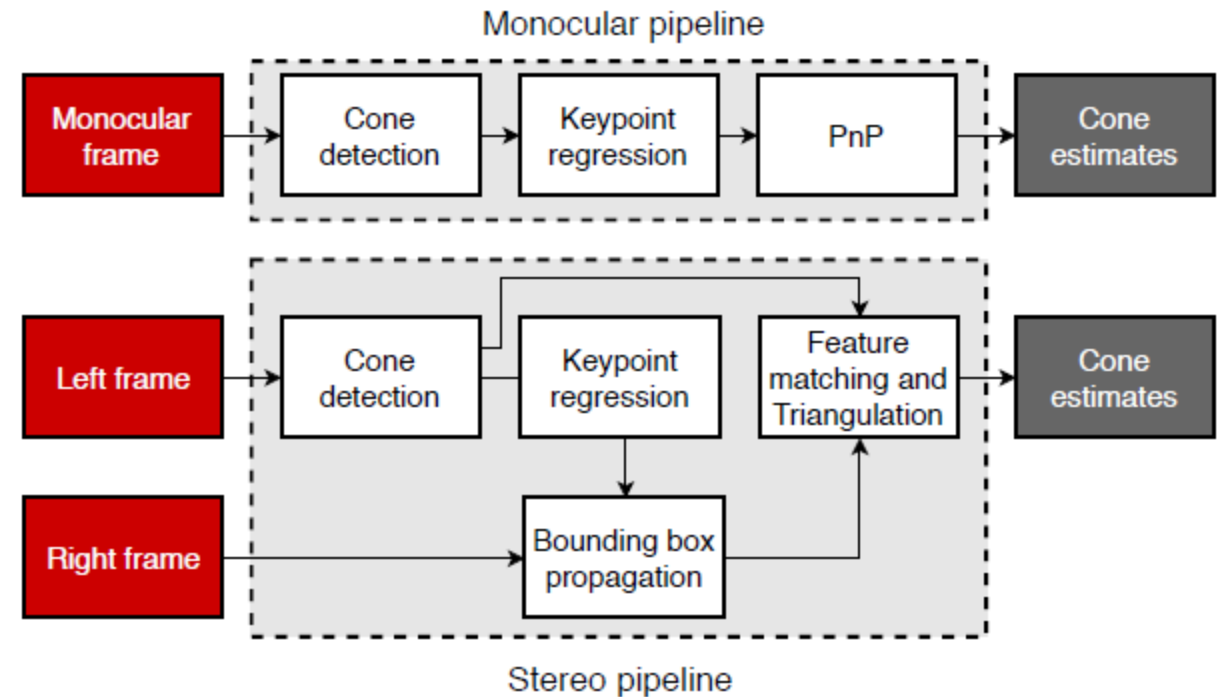
⚠ MOTION DISTORTION

- A scan isn't instantaneous
- Car moves mid-sweep → smeared points
- Can fabricate "ghost cones" if ignored
- Fix: undistort using velocity estimate

Camera Pipeline: Mono vs. Stereo

PERCEPTION · CAMERA

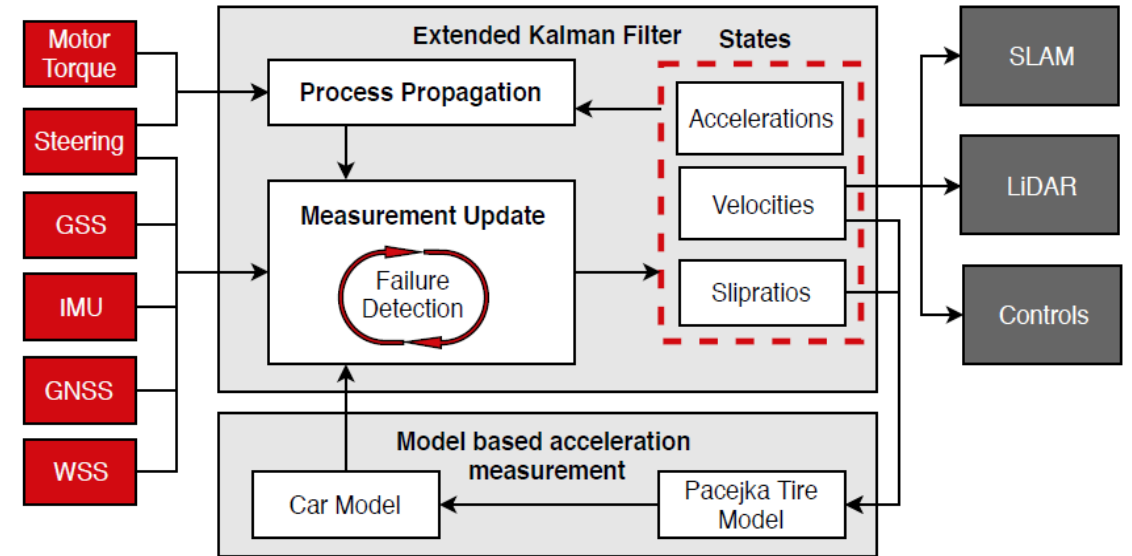
- YOLOv2 detects + classifies cones
- CNN regresses 7 keypoints per cone (corner-like features)
- Mono: keypoints + known geometry $\rightarrow$ PnP $\rightarrow$ 3D pose
- Stereo: propagate bbox to right frame $\rightarrow$ match $\rightarrow$ triangulate



Velocity Estimation

VELOCITY EST.

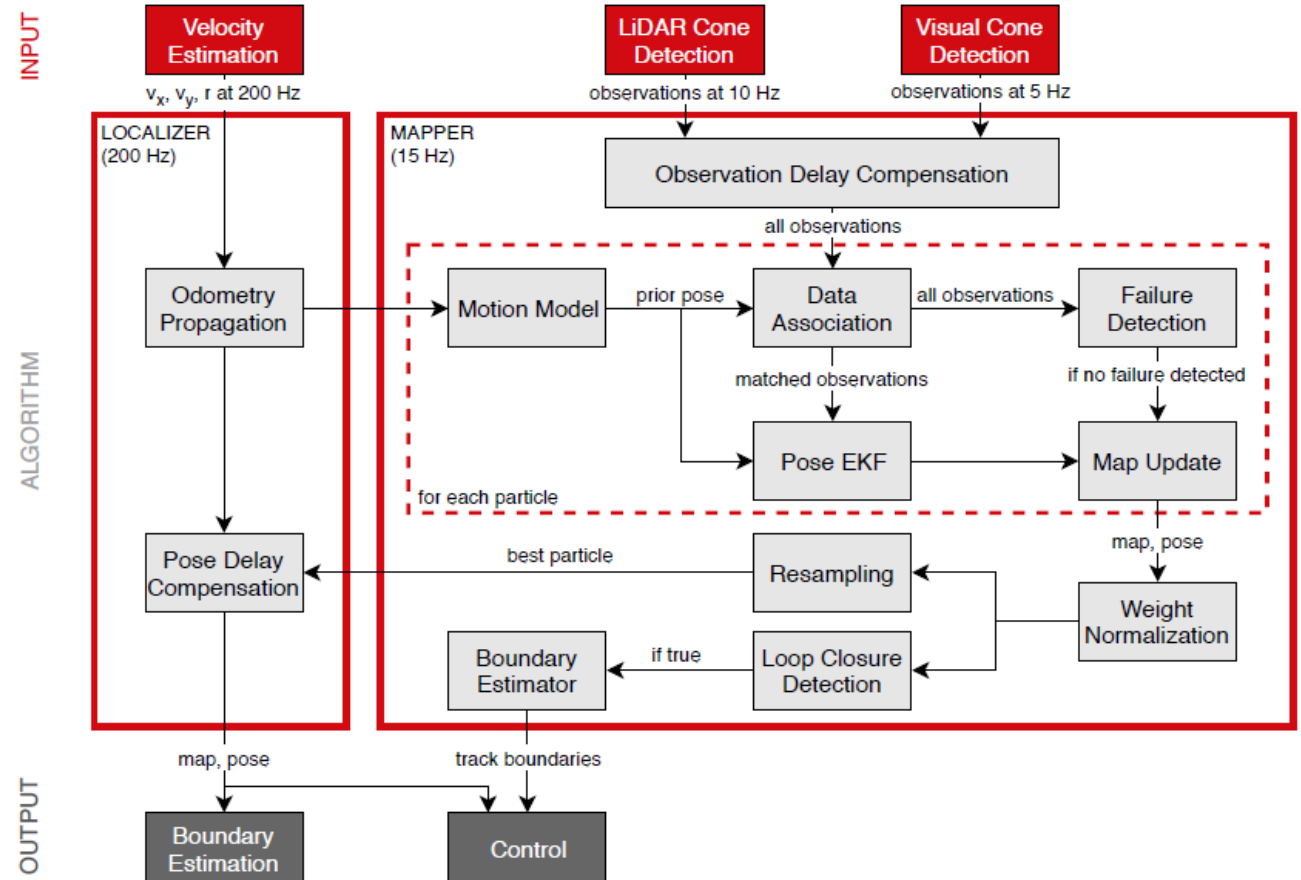
- Why: feeds LiDAR undistortion, SLAM motion model, control
- 9-state EKF fuses 6 sensors (motor torque, steering, GSS, IMU, GNSS, WSS)
- Why fuse, not trust one sensor: each has own bias/noise/failure mode
- Outliers (chi-squared, instant) vs. drift (variance-based, slow bias)



SLAM & Mapping

SLAM

- Why: track unknown $\rightarrow$ must map + localize together
- Core difficulty: data association (which cone is which?)
- FastSLAM 2.0 — particle filter, multi-hypothesis, scales linearly
- Lap closure $\rightarrow$ map frozen $\rightarrow$ switch to Localization mode



Control: Bicycle Model + MPC

CONTROL

- Why not just track center line: racing $\neq$ minimum tracking error
- Bicycle model — dynamic (tire forces) blended w/ kinematic (low speed)
- Contouring MPC — maximizes progress along path over a horizon
- Respects tire-friction + track-boundary constraints directly

Testing & Simulation

TESTING

- Why simulate first: real-world track time is scarce/costly
- FSSIM — first-principles model, simulated cone observations
- ATS — automated, distributed test runner (per-PR + nightly)